

PROJECT REPORT

Real-Time Public Transport Tracking for Small Cities

Academic Project Report

Theme: Smart Transportation & Urban Mobility

Domain: Artificial Intelligence / GPS & IoT / Web-Mobile Applications

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1. Executive Summary

Public transport in small and mid-sized Indian cities is largely bus-based, but commuters have no reliable way of knowing when the next bus will actually arrive. Printed timetables rarely reflect real road conditions, and the absence of live information forces commuters to wait at stops for unpredictable periods, discourages first-time riders, and pushes many towards private vehicles or unregulated para-transit. This report presents the design and implementation plan for a Real-Time Public Transport Tracking system that uses GPS-enabled devices on buses to provide commuters with live vehicle location and accurate arrival predictions, built specifically for the scale and budget constraints of smaller cities.

The proposed system consists of three integrated components: a lightweight GPS Tracking Module fitted on each bus that periodically reports its location, a Transit Authority Dashboard that lets the local transport department monitor fleet position, route adherence, and delays, and a Commuter-facing mobile/web application that shows live bus location on a map along with estimated arrival time at the user's stop. The solution is designed to work with low-cost hardware and standard smartphones, keeping deployment realistic for a small-city transit budget.

This report documents the problem context, proposed architecture, technology stack, implementation methodology, feasibility analysis, and expected impact of the system, and is intended to serve as both a technical blueprint and an academic submission reference.

1.1 Key Highlights

- Real-time GPS location reporting from each bus at short, configurable intervals.
- Live map view for commuters showing all active buses on their route with estimated arrival time.
- Delay and route-deviation alerts for the transit authority based on scheduled vs actual position.
- ETA prediction that combines live GPS speed with historical travel-time patterns per route segment.
- Centralised dashboard for the local transport department to monitor the entire fleet from one console.
- Designed around low-cost GPS/GSM hardware and standard smartphones, keeping cost realistic for small-city budgets.

2. Problem Statement

2.1 Problem Definition

In most small and mid-sized Indian cities, public bus services operate without any live tracking system. Commuters rely on fixed printed schedules or word-of-mouth estimates that rarely match actual arrival times, since buses are affected by traffic, road conditions, and driver behaviour. This uncertainty leads to long, unproductive waiting at stops, missed buses, and a general loss of confidence in public transport as a dependable option. There is a need for an affordable, easy-to-deploy tracking system that gives commuters accurate, real-time information without requiring the scale of infrastructure used by large metro transit authorities.

2.2 Background and Context

Large metropolitan cities have deployed sophisticated Intelligent Transport Systems (ITS) with dedicated control rooms, extensive sensor networks, and integrated fare systems. These systems are effective but expensive, and their scale is not appropriate for smaller city transit fleets that may operate only a few dozen buses across a handful of routes. At the same time, GPS-enabled tracking hardware and mobile connectivity have become inexpensive and widely available, making it feasible to build a scaled-down, focused tracking solution suited to smaller fleets without the overhead of a full ITS deployment.

2.3 Stakeholders Affected

Stakeholder	Pain Point
Commuters	Unpredictable wait times at bus stops, no visibility into bus location or delays.
Bus Drivers / Conductors	No systematic way to report delays or route deviations to the depot.
Transit Authority Staff	Limited real-time visibility into fleet position, punctuality, and route adherence.
Local Municipal Body	Lack of data-driven insight into service reliability for planning and route optimisation.

2.4 Scope of the Problem

The scope of this project is limited to live GPS-based bus tracking, ETA prediction for commuters, and a monitoring dashboard for the transit authority. It does not, in its first phase, address digital ticketing/fare collection or multi-modal journey planning, though the architecture is designed to support these as future extensions.

3. Objectives

The project is guided by the following core objectives, each mapped to a measurable outcome:

Objective	Measurable Outcome
Track live location of every bus in the fleet	Location updated every 10-15 seconds from each bus's GPS device
Provide accurate arrival predictions to commuters	ETA displayed per stop, refined using live speed and historical data
Detect and alert on route deviation or excessive delay	Alert raised within a few minutes of deviation from expected path/schedule
Reduce commuter waiting uncertainty	Measurable reduction in perceived/actual wait time vs no-tracking baseline
Provide centralised fleet monitoring	Dashboard showing live status of all active buses and routes
Keep deployment cost low	Use of low-cost GPS/GSM trackers and standard smartphones for the commuter app

4. Existing Solutions and Comparative Analysis

Several public transport tracking approaches currently exist in India and globally. Reviewing their strengths and limitations helped shape the design priorities of this project.

4.1 Comparative Overview

Solution	Strengths	Limitations for Smaller Deployments
Metro city ITS platforms (e.g. Chalo, ITS control rooms)	Mature, feature-rich, integrated with fare and control systems	High cost, complex integration, built for very large fleets
Printed timetables (status quo)	Zero technology cost, simple to distribute	Does not reflect real-time conditions, frequently inaccurate
Generic ride-sharing/maps apps	Widely available, familiar interface	Not designed for fixed-route bus tracking or transit-authority monitoring
Manual depot radio reporting	Low technology dependency, human judgement	Not scalable, no data logging, no commuter-facing output

4.2 Identified Gap

Existing large-scale ITS platforms assume fleet sizes and budgets that small-city transit departments do not have, while the status quo of printed timetables provides no real-time value at all. The gap this project addresses is a lightweight, low-cost GPS tracking and ETA system purpose-built for small fleets, prioritising ease of deployment and affordability over the extensive feature set of metro-scale systems.

5. Proposed Solution

The proposed system has three integrated components: a GPS Tracking Module fitted on each bus, a Transit Authority Dashboard, and a Commuter-facing tracking application, connected through a backend that manages live location data, ETA computation, and analytics.

5.1 GPS Tracking Module (On-Bus)

- Low-cost GPS/GSM tracking device fitted on each bus, reporting location at short fixed intervals.
- Reports coordinates, speed, and heading to the backend over the mobile network.
- Optionally uses the driver's/conductor's smartphone as a fallback tracking device where dedicated hardware is not yet installed.
- Buffers location data locally during signal loss and syncs once connectivity resumes.

5.2 Transit Authority Dashboard

- Displays live position of every active bus on a route map.
- Flags buses running significantly behind schedule or deviating from the expected route.
- Shows historical punctuality and route-adherence trends by route and time of day.
- Sends alerts for tracking-device failure or prolonged loss of signal from a bus.

5.3 Commuter-Facing Application

- Shows live bus location on a map for the commuter's selected route or stop.
- Displays estimated arrival time at the commuter's chosen stop, updated continuously.
- Lets commuters browse routes and stops without account creation for quick, guest-first access.

5.4 Unique Selling Points

- Built around low-cost GPS/GSM hardware, keeping per-bus cost low for small transit fleets.
- Smartphone-based fallback tracking allows a pilot rollout before dedicated hardware is procured.
- ETA model blends live speed with historical segment travel times for realistic predictions.
- Modular rollout — can be piloted on a single route before fleet-wide expansion.
- Simple, guest-first commuter app designed for quick adoption without onboarding friction.

6. System Architecture

6.1 High-Level Architecture

The system follows a layered architecture: on-bus tracking devices, a backend API and real-time communication layer, a database layer, and client applications for the transit authority and commuters.

Layer	Component	Technology
Edge	GPS Tracking Device	GPS/GSM tracker module or driver-side smartphone app
Edge	Location Reporting	HTTP/MQTT push to backend at fixed intervals
Server	REST + WebSocket API	Node.js, Express.js, Socket.IO
Server	ETA Prediction Engine	Python service (speed + historical segment timing logic)
Data	Primary Database	MongoDB / PostgreSQL with geospatial indexing
Client	Authority Dashboard	React.js
Client	Commuter App	React.js (Progressive Web App)
External	Maps & Geolocation	OpenStreetMap + Leaflet / Google Maps API
Infra	Hosting	Cloud VM / PaaS

6.2 Data Flow Overview

Each bus's tracking device reports its GPS coordinates and speed to the backend at short fixed intervals. The backend feeds this location data into the ETA prediction engine, which computes updated arrival estimates for upcoming stops on the route, and simultaneously stores the location for historical analytics. The current bus position and ETA are broadcast in real time to both the transit authority dashboard and the commuter app.

6.3 Component Interaction Diagram (Description)

Bus GPS Device -> Backend Location Ingestion -> ETA Prediction Engine -> Database (historical log) and -> Real-Time Broadcast Service -> Authority Dashboard and Commuter App.

7. Technology Stack — Detailed Rationale

7.1 Location Tracking

A GPS/GSM tracker module was selected for on-bus deployment because it provides continuous location reporting independent of a driver's phone, while a smartphone-based fallback keeps early pilot deployment low-cost. Both approaches report coordinates and speed over the mobile network at short intervals.

7.2 Backend

Node.js with Express.js provides the REST API and coordinates with a Python-based ETA prediction microservice. Socket.IO enables real-time, bidirectional updates between buses and the dashboard/commuter clients.

7.3 Database

MongoDB (or PostgreSQL with a geospatial extension, depending on deployment preference) stores route configuration, live location snapshots, and historical logs used for punctuality analysis and ETA model refinement.

7.4 Frontend

React.js is used for both the authority dashboard and the commuter-facing Progressive Web App, allowing a single codebase pattern and a lightweight, installable experience for commuters without requiring an app-store download.

7.5 Summary Stack Table

Category	Technology Choice
Location Tracking	GPS/GSM tracker module, smartphone fallback (driver app)
Backend	Node.js, Express.js, Python (ETA engine)
Real-Time Layer	Socket.IO (WebSockets)
Database	MongoDB / PostgreSQL with geospatial indexing
Frontend (Dashboard & Commuter App)	React.js, Tailwind CSS
Maps	Leaflet.js / OpenStreetMap or Google Maps API
Hosting	Cloud VM / PaaS
Version Control	Git and GitHub

8. Detailed Feature List

8.1 GPS Tracking Module Features

Feature	Description	Priority
Live Location Reporting	Bus location reported every 10-15 seconds	Must-have
Route Deviation Detection	Flags buses moving off their assigned route path	Must-have
Delay Detection	Compares live position to scheduled timing	Must-have
Offline Buffering	Stores location locally during signal loss, syncs on reconnect	Should-have
Driver-App Fallback Mode	Uses driver's smartphone GPS where hardware is unavailable	Should-have

8.2 Authority Dashboard Features

Feature	Description	Priority
Live Fleet Map	Monitor all active buses and their current position	Must-have
Punctuality Analytics	Delay and route-adherence trends by route and time	Must-have
Deviation & Delay Alerts	Notify staff of route deviation or excessive delay	Should-have
Device Health Monitoring	Alerts on tracker failure or prolonged signal loss	Should-have

8.3 Commuter App Features

Feature	Description	Priority
Live Bus Map	Real-time position of buses on the selected route	Must-have
Stop-wise ETA	Estimated arrival time at the commuter's chosen stop	Must-have
Route & Stop Search	Browse available routes and stops	Should-have
Guest Access	Usable without account creation	Should-have

9. System Design and Database Schema

9.1 Core Data Entities

The system is built around four core entities: Routes, Buses, LocationLogs, and DelayEvents, each modelled as a database collection/table with relationships maintained through referenced IDs.

9.1.1 Route Schema

Field	Type	Description
_id	ObjectId	Unique identifier
name / stops	String / Array of GeoJSON	Route name and ordered list of stop coordinates
scheduledTimings	Array	Planned arrival time offsets per stop
status	String	active / suspended

9.1.2 Bus Schema

Field	Type	Description
_id	ObjectId	Unique identifier
registrationNo	String	Bus registration number
route	ObjectId (ref: Route)	Assigned route
currentLocation	GeoJSON Point	Latest reported position
status	String	active / delayed / offline

9.1.3 LocationLog Schema

Field	Type	Description
_id	ObjectId	Unique identifier
bus	ObjectId (ref: Bus)	Associated bus
coordinates	GeoJSON Point	Reported location
speed	Number	Reported speed at timestamp
timestamp	Date	Time of the location report

9.2 Indexing Considerations

LocationLog documents are indexed on bus and timestamp, with a geospatial index on coordinates, to support efficient time-range and proximity queries used for ETA computation and dashboard mapping.

10. API Design

The backend exposes a REST API for configuration and historical queries, and a WebSocket channel for real-time location and ETA updates.

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticate authority staff and issue JWT
GET	/api/routes	List all routes with stops and schedule
GET	/api/buses/:id	Get live status and history for a specific bus
GET	/api/routes/:id/eta	Get current ETA for all upcoming stops on a route
GET	/api/analytics/:routeId	Get historical punctuality analytics
WS	location:update	Bus device emits live GPS location
WS	eta:broadcast	Server pushes updated ETA to clients

11. Bus Arrival Prediction & Tracking Methodology

Computing an accurate arrival estimate from raw GPS data is central to the system's value. The project adopts a phased approach, balancing accuracy with implementation complexity.

11.1 Phase 1 — Live Speed Extrapolation Model

In the initial version, ETA for each upcoming stop is calculated by dividing the remaining route distance by the bus's current reported speed, with a minimum-speed floor to avoid unrealistic estimates during brief stops.

11.2 Phase 2 — Historical Segment Refinement

As the system accumulates location logs over time, ETA calculation is refined using historical average travel time for each route segment by time-of-day, accounting for predictable slowdowns such as peak-hour traffic or school timings.

11.3 Phase 3 — Predictive Delay Modelling (Future Scope)

A future enhancement could use a machine learning model trained on accumulated multi-route location logs to anticipate delays a few minutes in advance based on upstream traffic patterns, though this is out of scope for the initial deliverable.

12. UI/UX Design Principles

12.1 Design Philosophy

The dashboard and commuter app prioritise clarity over visual complexity, since transit authority staff need to interpret fleet status at a glance, and commuters need arrival information without any learning curve.

12.2 Key Design Decisions

- Map-first interface for both the dashboard and commuter app.
- Consistent colour coding: green for on-time, yellow for minor delay, red for significant delay.
- Prominent stop-wise ETA display on the commuter app home screen.
- Minimal data usage per screen load for commuters on limited mobile data.

13. Implementation Methodology

The project follows a phase-based development methodology suited to an academic project timeline, structured as an incremental, single-route-first build.

Phase	Duration	Deliverables
Phase 1: Research & Design	Week 1-2	Problem analysis, architecture design, schema design
Phase 2: GPS Tracking Module	Week 3-4	On-bus/device tracking integration and location reporting
Phase 3: ETA Prediction Engine	Week 5	Speed-to-ETA computation logic and backend API
Phase 4: Authority Dashboard	Week 6	Live fleet map, analytics, alerts
Phase 5: Commuter App	Week 7	Live map and stop-wise ETA display
Phase 6: Testing & Pilot	Week 8	Single-route simulation/pilot testing
Phase 7: Refinement	Week 9	Bug fixes, documentation, report finalisation

14. Testing Strategy

Testing Level	Focus Area	Tools/Approach
Unit Testing	ETA calculation logic, location parsing	Jest / PyTest
Accuracy Evaluation	ETA prediction error vs actual arrival	Logged trial-run comparison
Integration Testing	API endpoints and WebSocket event flow	Postman, Socket.IO test client
Simulation Testing	End-to-end behaviour under simulated GPS traces	Route-replay simulation scripts
Usability Testing	Commuter app and dashboard ease-of-use	Informal user testing

15. Feasibility Analysis

15.1 Technical Feasibility

All core components rely on mature, well-documented technologies (GPS/GSM modules, Node.js, React, standard databases). No specialised hardware beyond a low-cost tracker or smartphone is required, making implementation achievable within an academic project timeline.

15.2 Economic Feasibility

By using low-cost GPS/GSM trackers and, where needed, a smartphone-based fallback instead of proprietary ITS hardware, the solution keeps both capital and operating costs realistic for a small-city transit budget.

15.3 Operational Feasibility

The dashboard reuses familiar web-application interaction patterns, minimising the learning curve for transit authority staff, while the smartphone fallback mode ensures a pilot can begin even before dedicated tracking hardware is fully procured.

15.4 Legal and Regulatory Feasibility

The system processes vehicle location data rather than personally identifying commuter data. Standard data-protection practices — limited location-history retention, access-controlled dashboards, and encrypted data transmission — will be followed to align with applicable guidelines.

16. Impact Analysis

16.1 Social Impact

- Reduces commuter wait-time uncertainty and daily commute stress at bus stops.
- Improves accessibility of public transport information for first-time and occasional riders.

16.2 Economic Impact

- Encourages higher public-transport ridership by improving reliability perception, reducing dependence on private vehicles.
- Lower deployment cost compared to full metro-scale ITS makes the solution replicable across smaller cities.

16.3 Environmental Impact

- Higher public-transport adoption driven by reliable tracking can reduce per-capita private-vehicle usage and emissions.

16.4 Governance Impact

- Provides transit authorities with data-driven insight into route punctuality for evidence-based scheduling.

17. Risk Analysis and Mitigation

Risk	Impact	Mitigation Strategy
GPS signal loss in dense urban areas	Inaccurate or missing location	Offline buffering and last-known-position fallback display
Tracking device failure	Bus disappears from live map	Automatic alert to authority and smartphone fallback mode
Inaccurate ETA during unusual traffic	Commuter frustration from mismatched predictions	Continuous refinement using historical segment data
Low smartphone/app adoption by commuters	Limited system impact	Guest-first, no-signup access to reduce adoption friction
Budget constraints for fleet-wide scaling	Slow rollout	Phased, route-by-route deployment starting with a pilot

18. Cost Estimation

The following is an indicative cost estimate for a pilot deployment covering 5-10 buses on 1-2 routes, intended to guide budget planning.

Cost Component	Estimated Cost (INR/year)	Notes
GPS/GSM Tracker Devices	3,000 - 6,000 per bus	One-time hardware cost, plus SIM data charges
Cloud Hosting (Backend + Database)	15,000 - 30,000	Scales with number of buses and routes
SIM/Data Connectivity	1,000 - 2,000 per bus	Recurring annual connectivity cost
Maintenance and Support	20,000 - 40,000	Ongoing app updates and bug fixes

19. Future Scope

- Digital ticketing and fare-collection integration using the same commuter app.
- Multi-modal journey planning combining bus routes with para-transit and walking segments.
- Machine-learning-based delay prediction using aggregated multi-route historical data.
- Crowding-level estimation per bus using onboard passenger-count sensors.
- SMS/IVR-based ETA access for commuters without smartphones.

20. Conclusion

The Real-Time Public Transport Tracking system proposed in this report directly addresses the unpredictability of bus arrival times in small cities by introducing affordable, GPS-based live tracking and ETA prediction. By relying on low-cost tracking hardware and a smartphone-based fallback, the solution balances technical soundness, economic feasibility, and operational practicality — qualities essential for adoption by small-city transit authorities with limited budgets.

The phased implementation plan, starting with a single-route pilot and progressively scaling to the full fleet, ensures that the system can be validated and refined based on real-world route behaviour before wider deployment. This report has laid out the complete technical and strategic foundation for the project — from problem framing through architecture, implementation methodology, feasibility, and long-term impact.

21. References

1. Ministry of Housing and Urban Affairs, Government of India — Urban Transport Guidelines.
2. GPS/GSM Tracking Module Technical Documentation (generic industry reference).
3. OpenStreetMap and Leaflet.js Documentation.
4. Socket.IO Documentation — Real-time bidirectional event-based communication.

- 5. React.js Official Documentation.
- 6. MongoDB Documentation — Geospatial Indexing.
- 7. Node.js and Express.js Official Documentation.

22. Appendices

A. Glossary of Terms

Term	Definition
GPS	Global Positioning System
GSM	Global System for Mobile Communications, used for data transmission from trackers
ETA	Estimated Time of Arrival
Geospatial Index	A database index optimised for location-based queries
WebSocket	A communication protocol enabling real-time, two-way data exchange

B. Team and Contribution

This project was developed by C.DARSHAN as a solo participant, encompassing problem research, system design, and full-stack implementation planning for the Real-Time Public Transport Tracking for Small Cities system.

C. Document Revision Notes

This report represents the complete project documentation prepared for academic submission, covering problem analysis through implementation methodology, feasibility, and future scope.